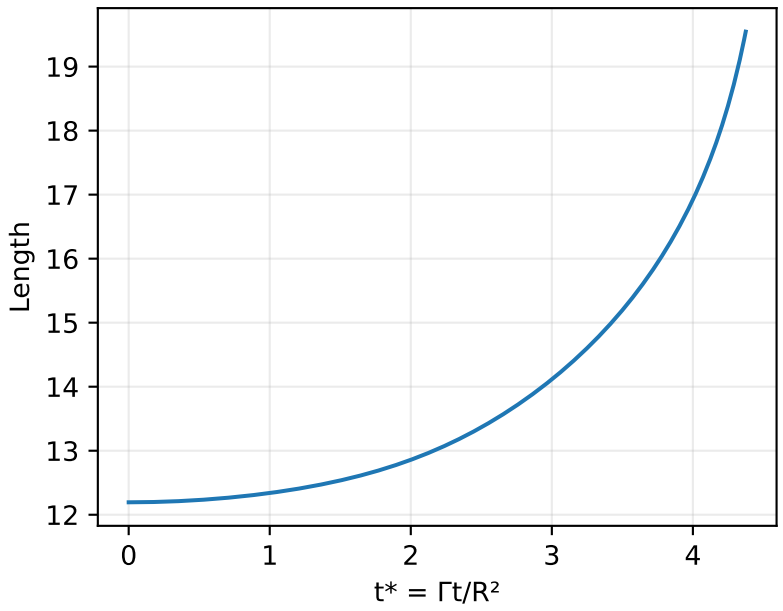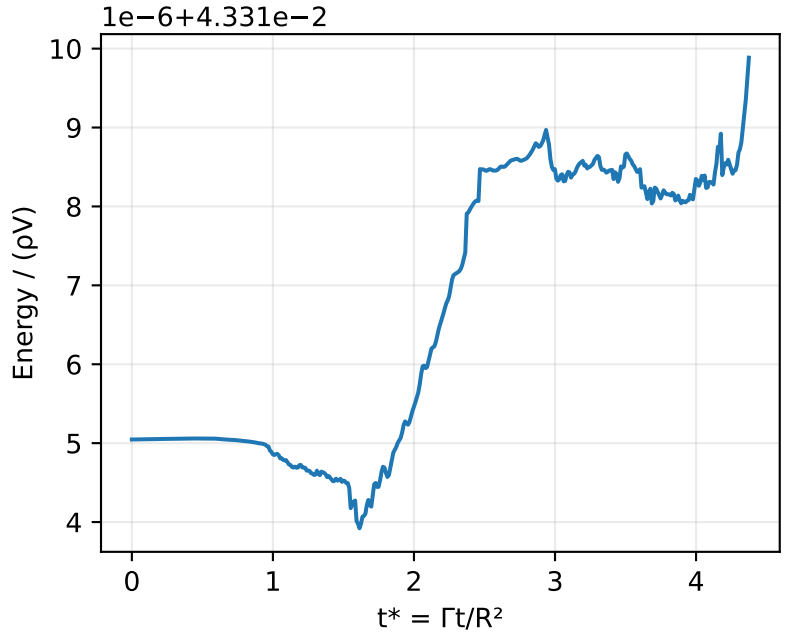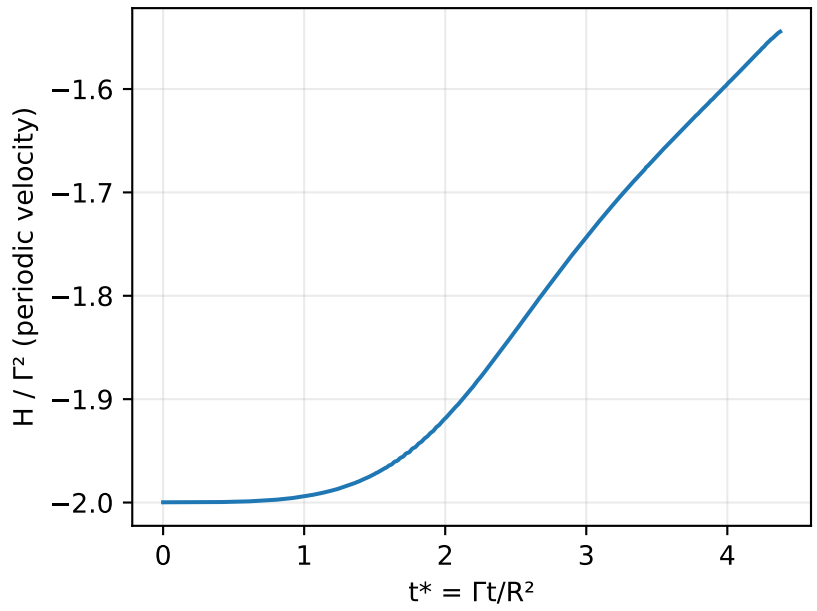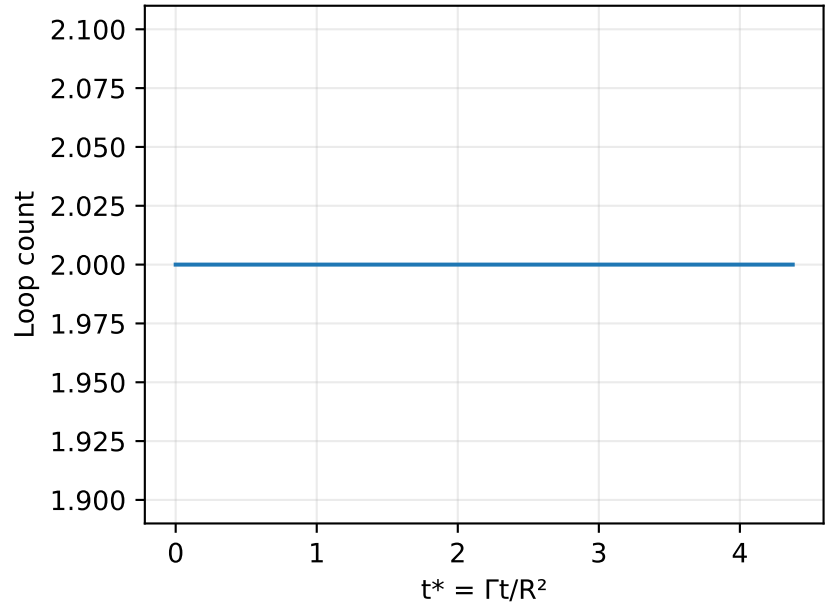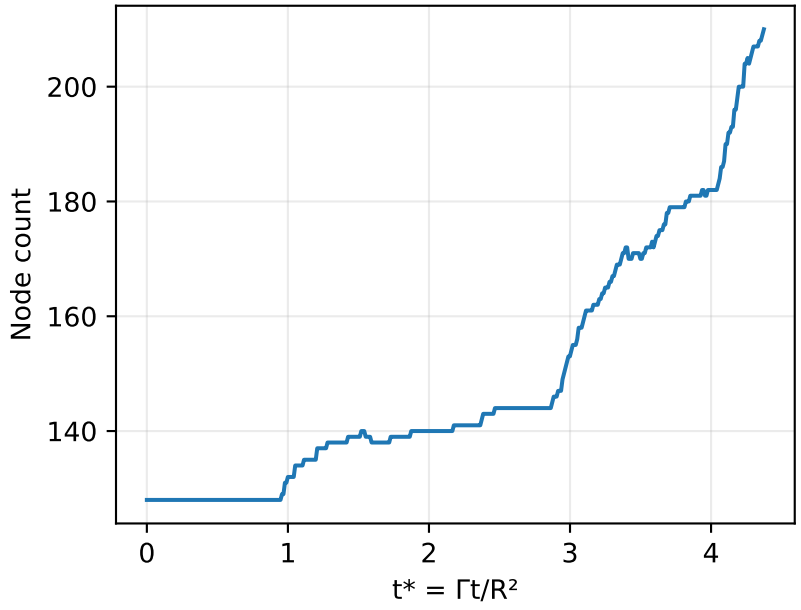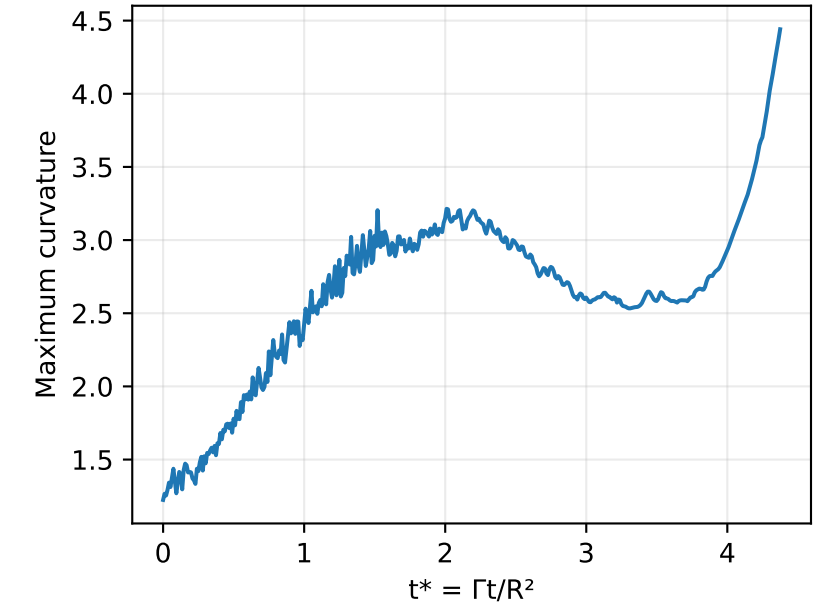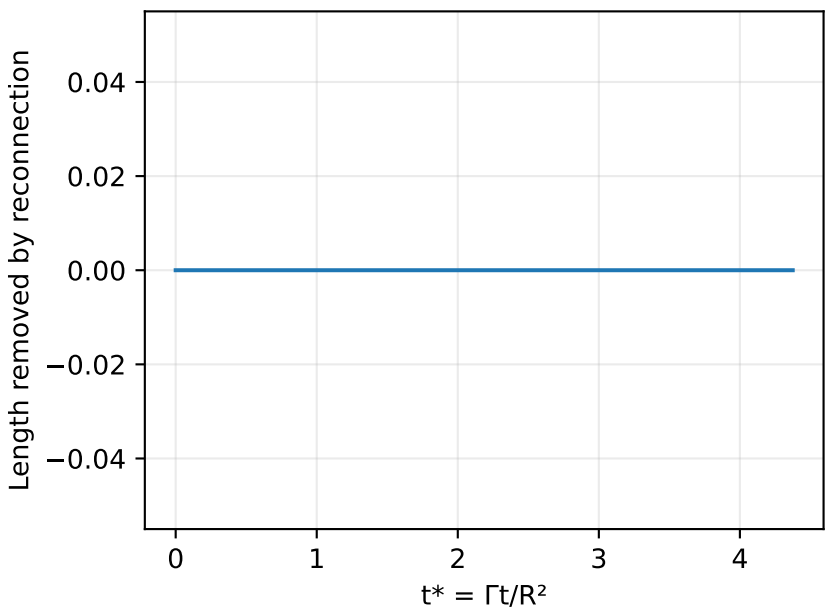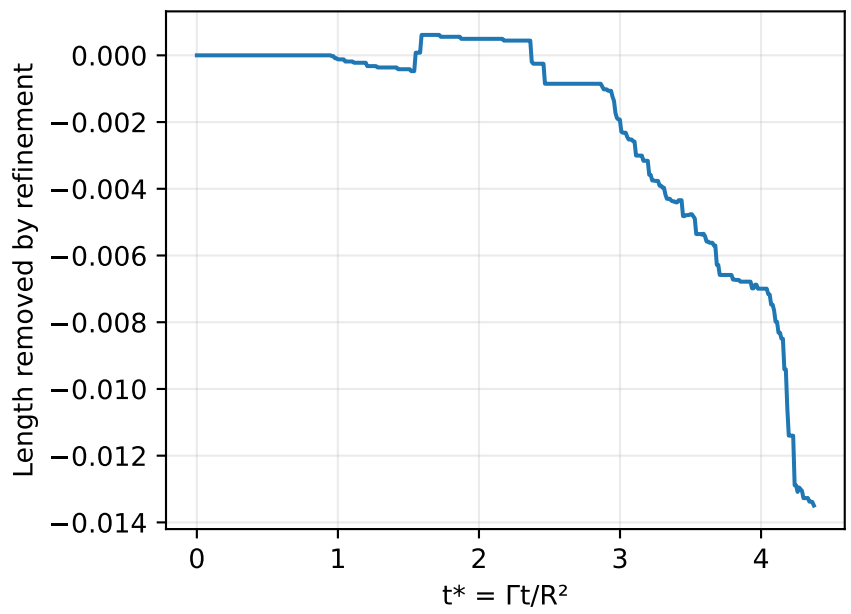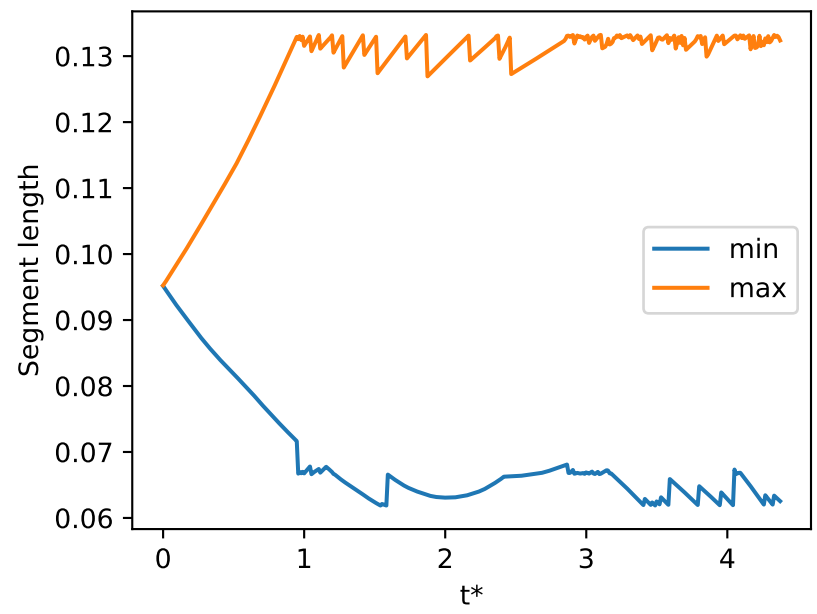hopf_prepared_halfdt · a=0.1171875 · no mutual friction